[SYSTEM STATUS: ONLINE]
// ROCKY LINUX 9.8 // SLURM SCHEDULER

UC Merced High-Performance Computing

The Researcher's Onboarding Playbook

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Facility Blueprint: The Borg Cube

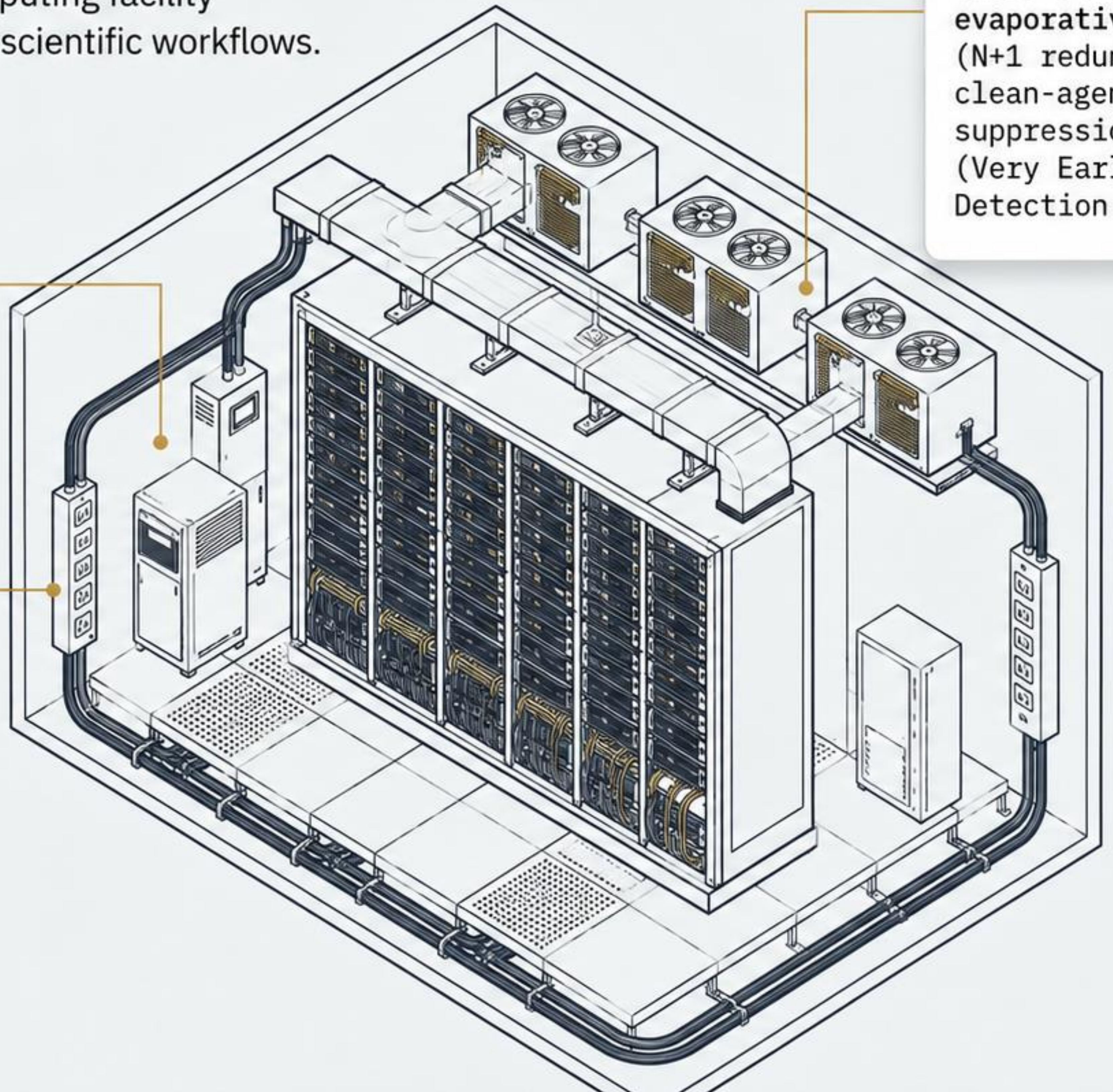
A 1,448-square-foot modular research computing facility engineered for uninterrupted, high-capacity scientific workflows.

Capacity

Single large-capacity rack (Up to 1,020U total capacity).

Power Redundancy

400 kW of N+1 redundant power. Dual 500 kVA PDUs and UPS units ensure zero-downtime operations.

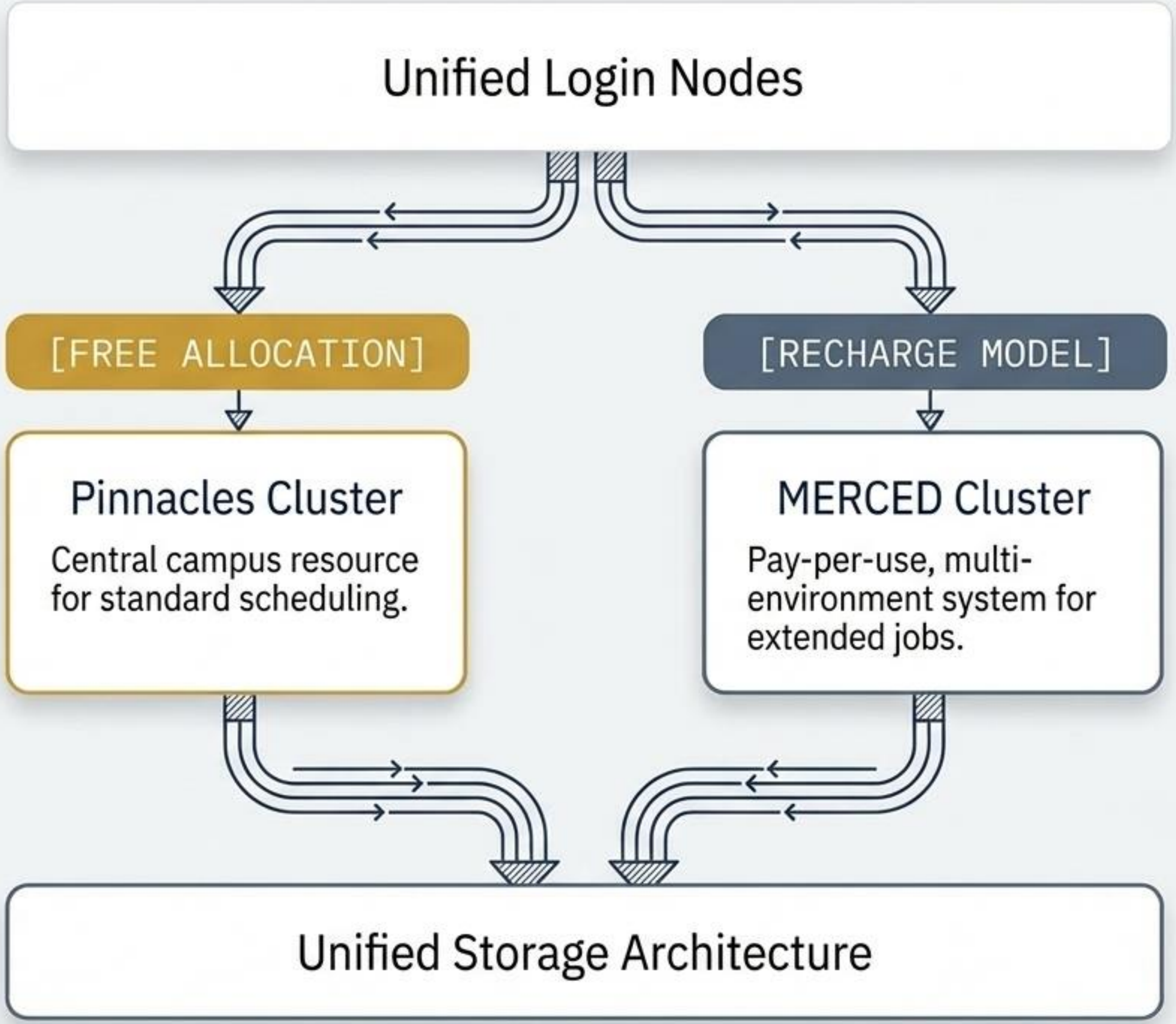


Environmental Safeguards

Three 60-ton indirect **evaporative cooling units** (N+1 redundancy), clean-agent FM-200 fire suppression, and VESDA (Very Early Smoke Detection Apparatus).

The Dual-Cluster Ecosystem

UC Merced operates a centralized high-performance computing environment. A single unified login grants you access to both shared storage and two distinct computational clusters—each designed for different research strategies.



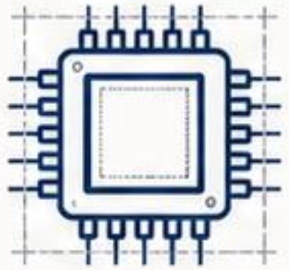
Cluster Diagnostic Matrix

Choose your computational environment based on your job length, budget, and urgency.

Criteria	Pinnacles	MERCED
Financial Model	Free to all UCM Faculty/PIs	Recharge: $0.10 per core-hour
Wall Time Limits	Strict queue limits, max 3 days	**UNLIMITED - run exceptionally long jobs without constraints**
Queue Wait Times	Standard congestion	Shorter waits - less crowded queues
Infrastructure Strategy	Centrally funded campus baseline	Recharge funds 100% applied to network/hardware maintenance; zero profit for CIRT

Pinnacles Hardware Profiles

High-performance computing environment interconnected with 100 Gb/s HDR InfiniBand and managed by Slurm.

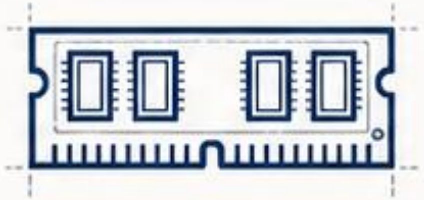


Standard CPU

49 Compute Nodes

40 nodes
2 × Intel Xeon Gold 6330
28 cores per CPU • 56 cores/node
256 GB RAM

9 nodes
2 × Intel Xeon Gold 6530
32 cores per CPU • 64 cores/node
256 GB RAM



High-Memory CPU

8 High-Memory Nodes

4 nodes
2 × Intel Xeon Gold 6330
28 cores per CPU • 56 cores/node
1 TB RAM

4 nodes
2 × Intel Xeon Gold 6530
32 cores per CPU • 64 cores/node
1 TB RAM

GPU Nodes

20 GPU Nodes / 40 GPUs

8 nodes • 2 × NVIDIA A100 PCIe
40 GB HBM2 per GPU

8 nodes • 2 × NVIDIA L40S
48 GB GDDR6 per GPU

4 nodes • 2 × NVIDIA H200 NVL
141 GB HBM3 per GPU

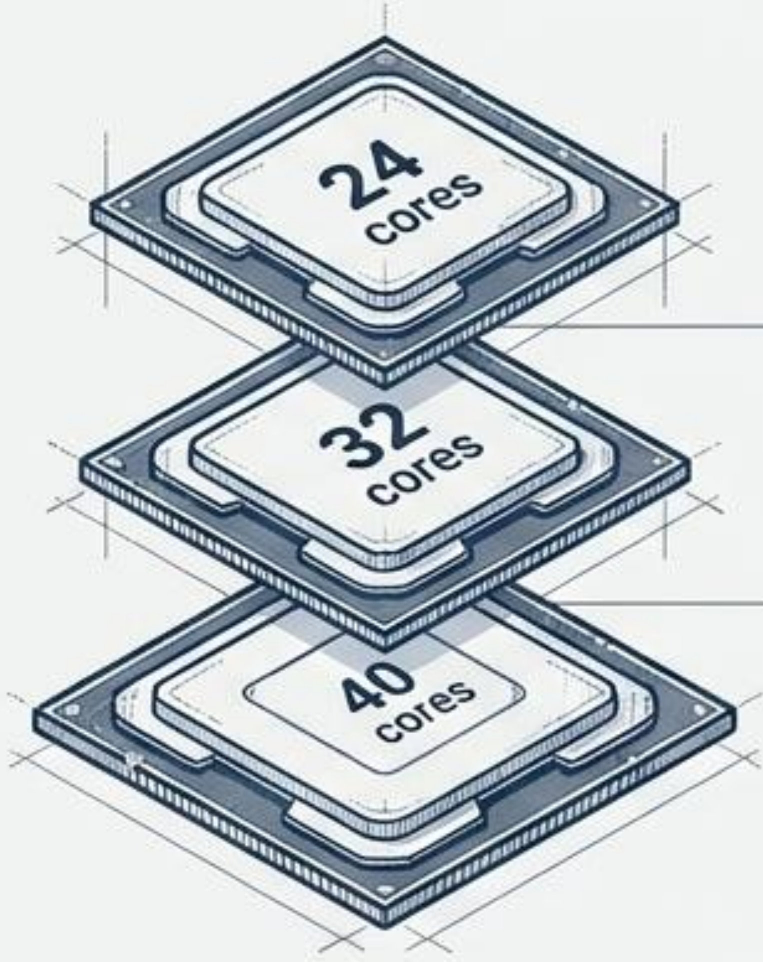
CENVAL-ARC NSF Resources

NSF Award #2346744 expands Pinnacles capacity with dedicated CENVAL-ARC resources and OSG-accessible test nodes.

MERCED Hardware & Recharge Math

MERCED (Multi-Environment Research Computer for Exploration and Discovery) provides 1,872 CPU cores across 66 nodes (including 25 high-memory nodes).

Multi-Generational Nodes



Broadwell Architecture:
24 cores/node
(112GB - 257GB RAM)

Skylake Architecture:
32 cores/node
(191GB RAM)

Cascadelake Architecture:
40 cores/node
(191GB RAM)

Expect relative performance variations depending on which node your job hits.

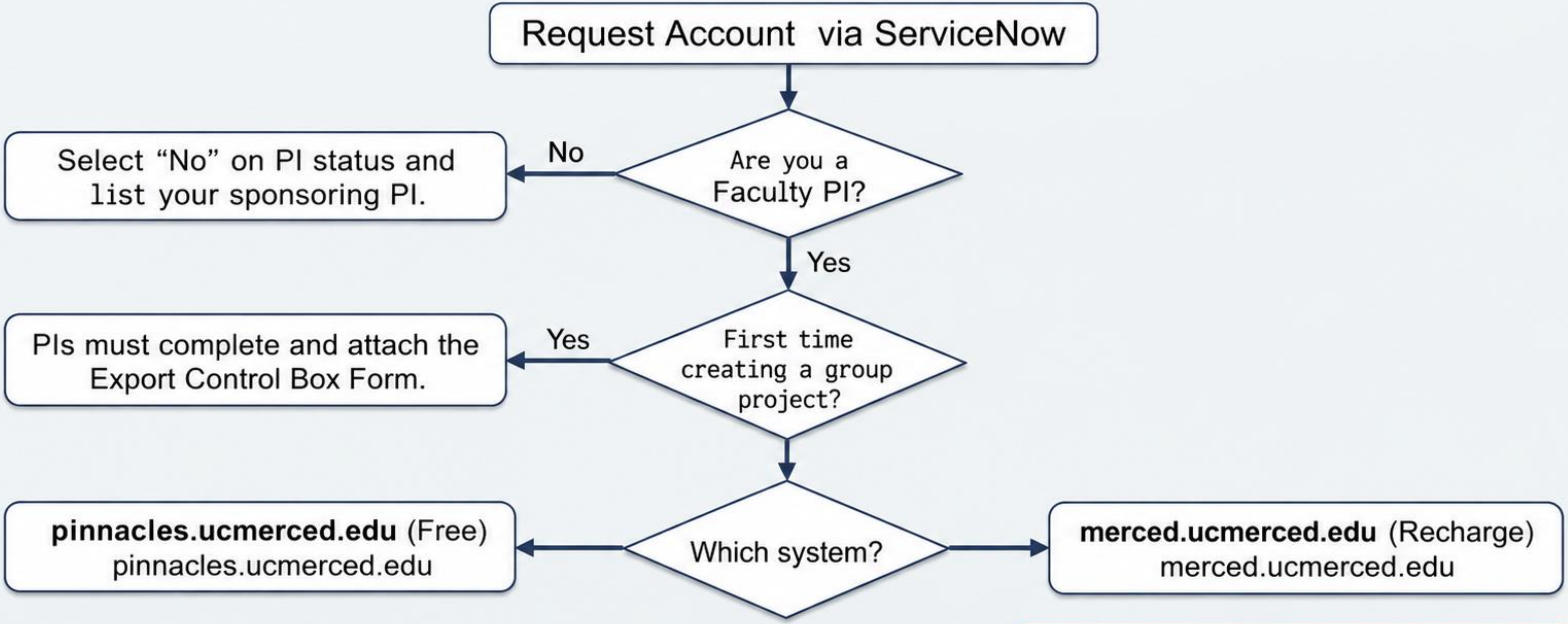
Recharge Cost Calculator

$$\textbf{TOTAL COST} = \text{\# of Cores} \times \text{Wall Clock Hours} \times \$0.10$$

- Definition: One “core-hour” = 1 compute core + 2GB RAM used for 1 hour.
- Requirement: A valid Chart of Account (COA) number is mandatory upon application.

The Access Pathway

All requests are routed through ServiceNow. Student accounts must be associated with a UC Merced PI.



CRITICAL: You must attach a valid **COA number** or the request will be automatically denied.

Centralized Entry Points

Access requires connection to the Campus VPN or on-campus eduroam/CatNet. A single login node grants access to both clusters.

Traditional Terminal (SSH & X11)

`ssh <UCMID>@login.rc.ucmerced.edu`

Graphics/X11: Use **`ssh -X`** (Mac requires XQuartz; Windows use MobaXterm's integrated X11 server).

Web GUI (Open OnDemand)

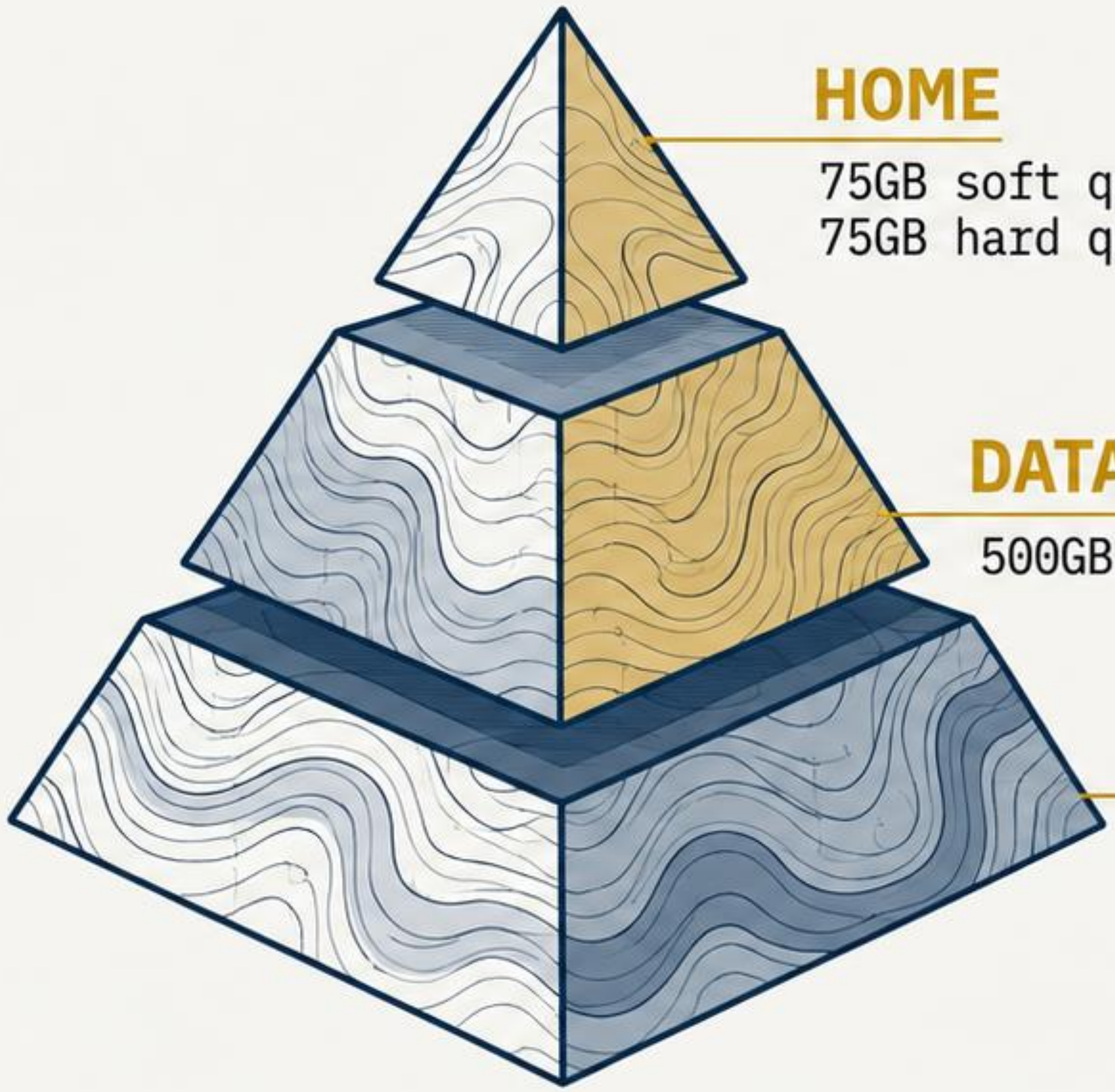
ondemand.rc.ucmerced.edu

Navigate to **`ondemand.rc.ucmerced.edu`** for a user-friendly, browser-based graphical interface.

DO NOT run computationally intensive processes on rclogin01-03. Login nodes are strictly for file prep, compiling, and job submission via queues.

Unified File System Architecture

Pinnacles and MERCED share a unified storage system over a 10G network. Users are responsible for their own data backups.



HOME

75GB soft quota / 75GB hard quota

DATA & SCRATCH

500GB soft quota each

BORGSTORE

Purchase rate $51.20/TB/yr

System Rules Widget

- **Purge Warning:** Scratch is periodically purged when total system capacity hits 85%. **DO NOT use for long-term retention**.
- **Temp Files:** Avoid writing to `/tmp` on head nodes; route temporary files to your personal scratch directory.
- **Grace Period:** 7-day grace period upon hitting soft quotas. Use `quota -vs` to monitor limits.

Global Module Environment

Pinnacles and MERCED utilize the Lmod system to manage software environments dynamically, preventing dependency conflicts.

```
module avail // Lists all available software modules

module load <mod_name> // Loads the specific software enviro

module list // Displays everything currently loaded

module unload <mod_name> // Removes the software environment

module swap <mod_1> <mod_2> // Swaps one loaded module for a

man module // Opens the complete manual for module commands
```

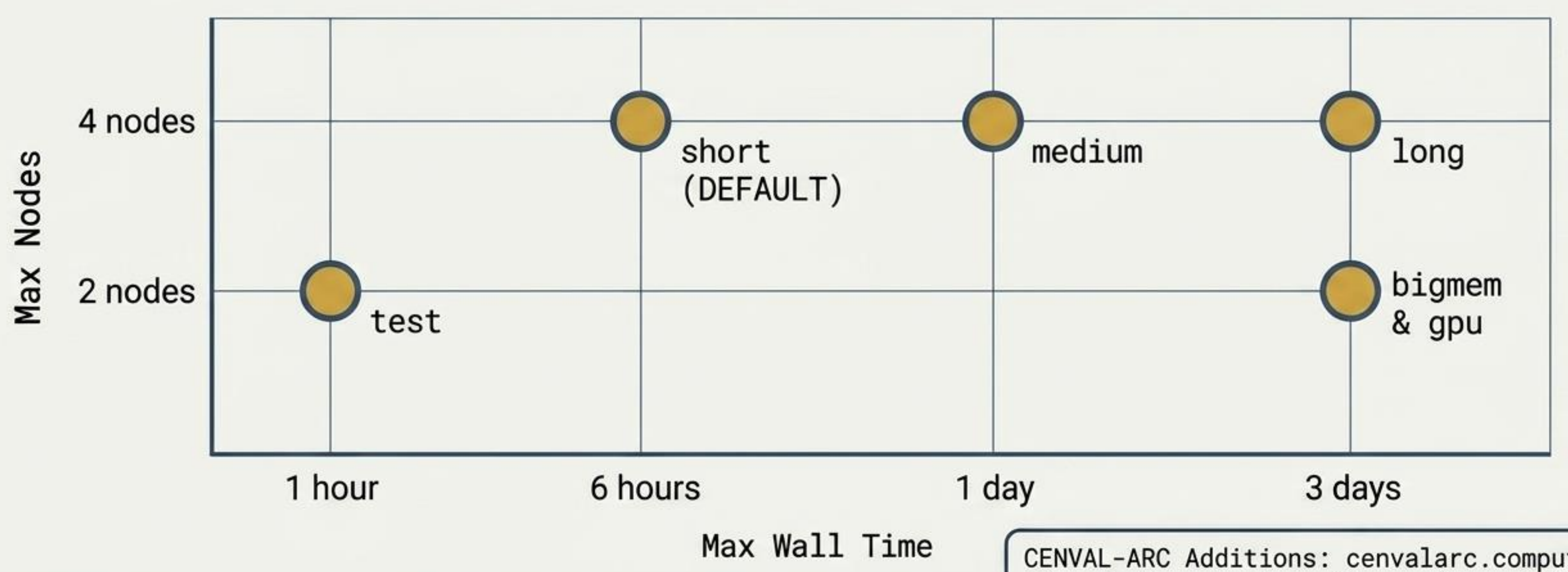
Pre-Installed Stack

Hundreds of modules are globally available. Popular packages include:

- Python / Anaconda
- R / Tidyverse
- MATLAB
- CUDA
- MPI
- Gromacs
- SLURM tools

Pinnacles Queue Strategy

Select the appropriate public queue based on your job's required time and node limits.



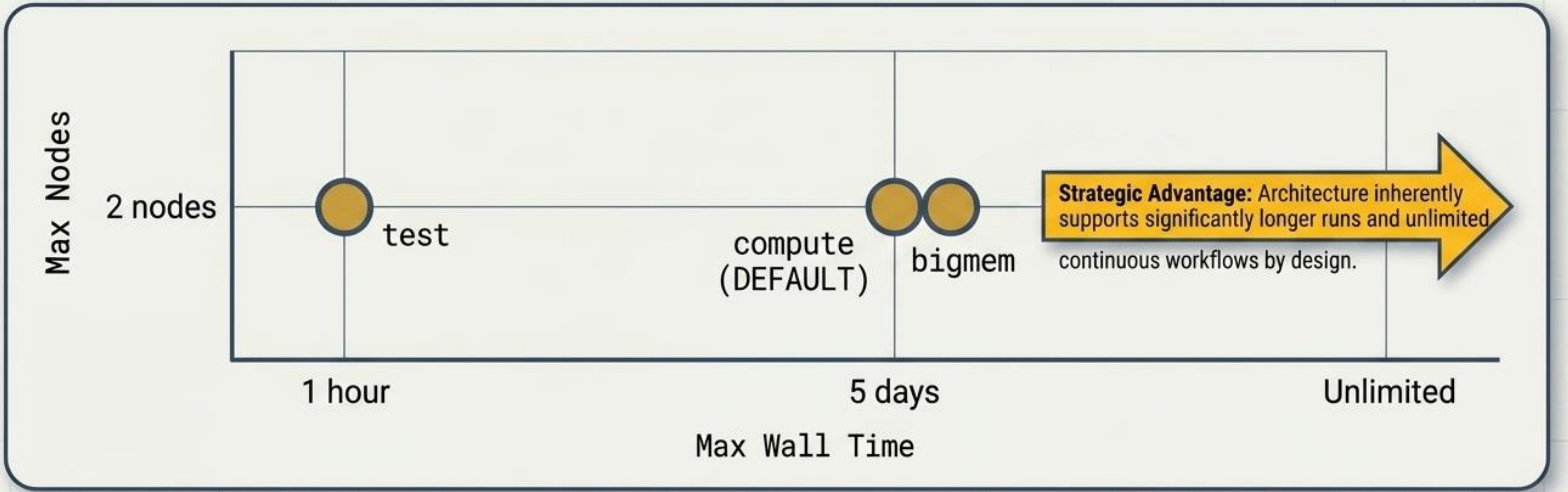
CENVAL-ARC Additions: cenvalarc.compute (4 nodes), cenvalarc.bigmem (2 nodes), cenvalarc.gpu (2 nodes) - all capped at 3 days max wall time.

UC MERCED

MERCED Queue Strategy

The recharge environment operates distinct queues focused on long-running tasks.

SYNTAX ALERT: `#SBATCH -M merced` MUST always be used to route a job to this specific cluster.



The Code of Conduct: Dissemination & Citations

All users must formally acknowledge UC Merced OIT and CIRT in talks, posters, and manuscripts relying on HPC data.

```
Pinnacles Cluster
```

This research was conducted using the Pinnacles cluster, which is centrally funded by the University of California, Merced, and maintained by the Cyberinfrastructure and Research Technologies (CIRT) team at UC Merced.

```
CENVAL-ARC Addendum
```

...conducted using CENVAL-ARC compute resources on the Pinnacles cluster (NSF #2346744) at the Cyberinfrastructure and Research Technologies (CIRT) at University of California, Merced.

```
MERCED Cluster
```

...conducted using MERCED cluster, which is centrally funded by the University of California, Merced, and maintained by the Cyberinfrastructure and Research Technologies (CIRT) team at UC Merced.

UCM HPC/Jupyter Hub Documentation

The HPC Quick Reference

System Support (CIRT)

Ticketing: ServiceNow Hub (account requests & technical faults).

Live Support: Zoom Office Hours (Wednesdays & Thursdays 10:00am – 11:00am).

Community: Join the UC Merced HPC Cluster Slack Workspace (ucmhpcclusters.slack.com).

The 5 Essential Commands

```
ssh <UCMID>@login.rc.ucmerced.edu
 – Access system

quota -vs
 – Check data limits

du -h -s <directory>
 – Check folder size

module avail
 – List installable software

#SBATCH -M merced
 – Route job to Recharge cluster
```